



▽ Jokes Aside:
Measuring ×
the **Semantic Distance**
of Double Meanings ○



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RESEARCH QUESTION



Can we measure the
hilarity of a joke?

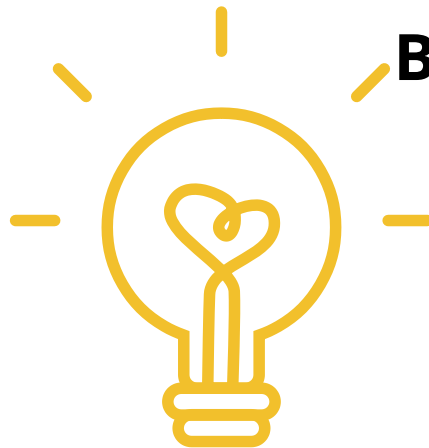


RESEARCH QUESTION

Note:

NOT if a joke is funny or not
BUT how funny is a joke on a scale 1-5

NOT is A funny or not ?
BUT is A funnier than B ?



Petrovic and Matthews (2013)
proposed **a model to generate jokes**

01

The “I like my X like I like my Y, Z” joke template

Example: I like my **ice** like I like my **dreams, crushed**

*The model, based on the incongruity theory, leveraged the frequency of co-occurrence of words in large corpus and was based on the assumption that **the degree of hilarity of a joke increases with:***

- *The increase of frequency of the use of the attribute contained in Z (“crushed”) to describe both X (“ice”) and Y (“dreams”).*
- *The decrease of the frequency of the general use of Z (“crushed”); in other words, its rarity.*
- *The increase of the degree of ambiguity of the attribute.*
- *The increase of the distance between X (“ice”) and Y (“dreams”).*





Results of the model

16.3%

Of the jokes generated by the model, 16.3% were considered funny by human raters

33%

Only 33% of jokes produced by humans using the same template that were considered funny by the same raters.

Winters et al. (2019) proposed **Goofer**: another model to generate jokes with the same template

02

01

Input a noun X

02

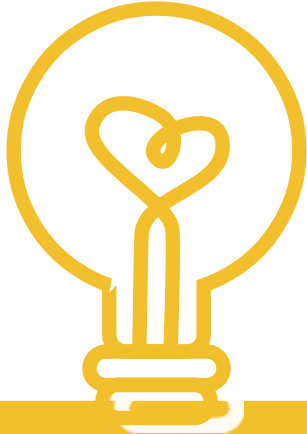
Generate a compatible adjective Z

03

Produce another noun Y that could be associated with Z

04

Apply a **rater model**: if the rate is above a certain threshold, publish the joke



— **The algorithm**



- Winters et al. (2019)

The metrics of the rater model



Obviousness	Refers to how natural and straightforward the initial interpretation appears.	Word Frequency Google Ngrams' <i>1-gram</i> data
Conflict	Arises when the punchline contradicts the initial meaning, triggering the need for reinterpretation.	Word Combination Frequency 2-grams Focusing on adjective-noun pairs The pairs were filtered using <i>WordNet</i> and a POS tagger to retain only relevant combinations.
Compatibility	Ensures the punchline aligns with a second, hidden meaning.	Word Combination Frequency + Number of Meanings: based on the number of definitions available for a word in <i>WordNet</i>
Comparison	Emphasizes the necessity for distinct interpretations.	Adjective Vector Similarity How similarly adjectives are distributed across nouns with <i>Word2Vec</i>
Inappropriateness 	Suggests that the second meaning should feel odd, inappropriate, or taboo, thereby enhancing the humorous effect.	Word Sexiness: Calculated by comparing word frequency in a sexual corpus (from textfiles.com) to its frequency in general usage



Winters et al. (2019)

Results of the model

11.4%

Of the jokes generated by the model, 11.4% were considered funny by human raters (compared to 16.3% of previous model)

27.4%

The previous model used a 3-point scale, while this model employed a more fine-grained 5-point scale. Under this new classification system, the percentage of human-generated jokes rated as funny also decreased, dropping to 27.4%.



Low hanging fruits ?

The idea

Convert the
metrics:
make use of
dynamic word
embeddings





Low hanging fruits ?

The idea

Convert the
metrics:
make use of
dynamic word
embeddings



Spoiler alert

**NO, NOT
REALLY**



This model

03

The “I like my **X** like I like my **Y**, **Z**” joke template

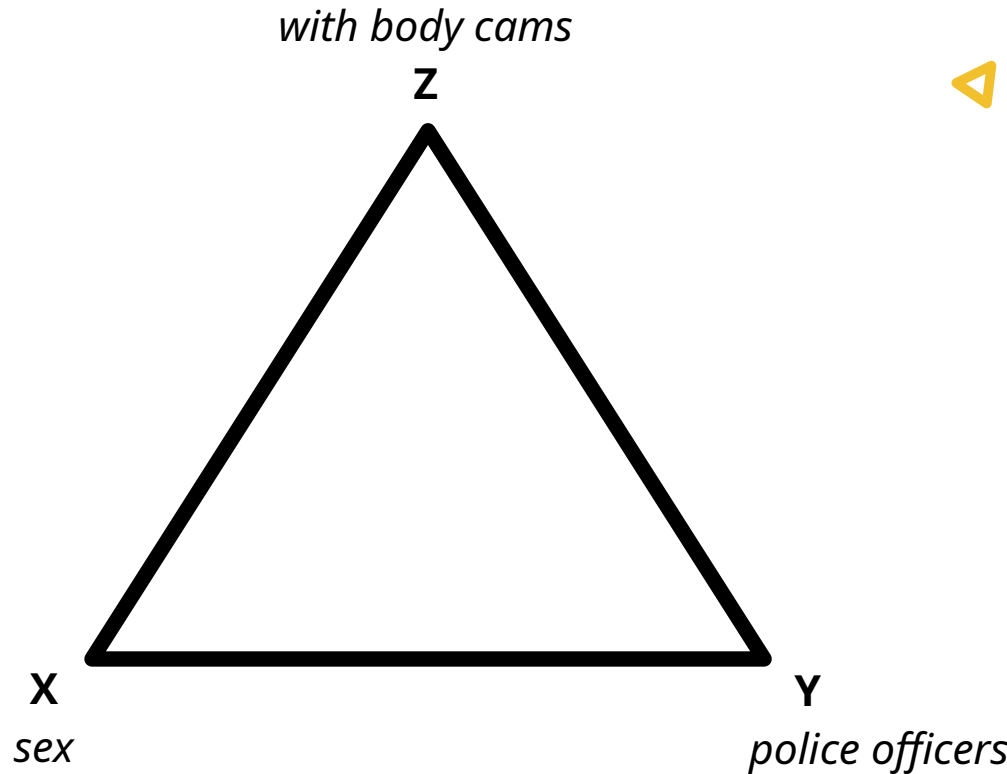
Example: I like my **sex** like I like my **police officers, with body cams**

Obviousness	Cosine similarity of the embedding vectors of X (“sex”) and Z (“with body cams”).	For the pun to work, this should be high .
Compatibility	Cosine similarity of the embedding vectors of Y (“police officers”) and Z (“with body cams”).	For the pun to work, this should be high .
Comparison	Cosine similarity of the embedding vectors of XZ (“sex with body cams”) and YZ (police officers with body cams”).	For the pun to work, this should be low .
Symmetry	Difference between obviousness and compatibility.	For the pun to work, this should be low .



The “I like my **X** like I like my **Y, Z**” joke template

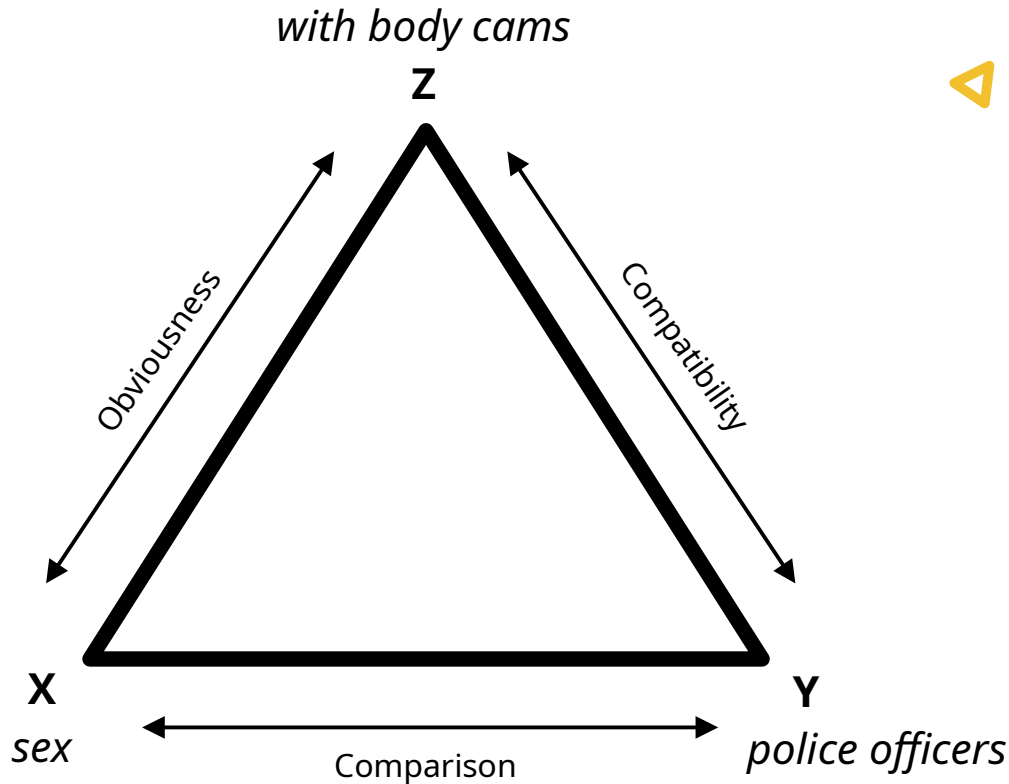
Example: I like my **sex** like I like my **police officers, with body cams**





The “I like my **X** like I like my **Y**, **Z**” joke template

Example: I like my **sex** like I like my **police officers, with body cams**



Symmetry

=

Obviousness

-

Compatibility



On Symmetry

(not only with “I like my X like I like my Y, Z” template)

The last metric addresses the question: **do the two meanings work equally well?**

Why did the photon check a bag at the airport? It didn't—it was traveling light!

The pun relies on the double meaning of the word “light,” which can be both a noun (representing the physical phenomenon that enables vision) and an adjective (meaning the opposite of heavy). The humor comes from the fact that the expression “traveling light” can be easily interpreted in both ways—its double meaning is symmetrical.

Would you like some soda in your whiskey? asked Tom caustically.

The play here is on “caustic soda,” but the problem is that “caustically” is not a common adverb used in reference to soda. To understand one side of the pun, a person must force their mind to connect two different forms to a single meaning, disrupting the symmetry and making the double meaning less effective.



TWO MODELS

OpenAI

text-embedding-3-small

MiniLM (Microsoft)

all-MiniLM-L6-v2



THREE DATASETS

JokeJugder

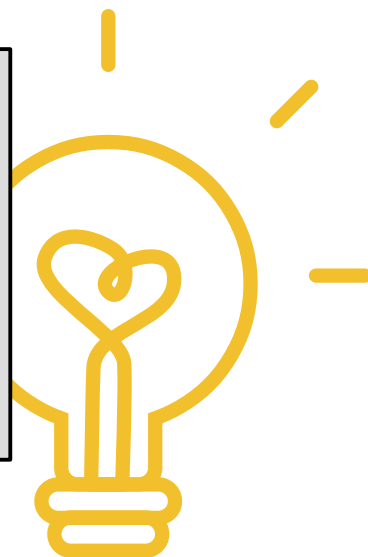
Winters *et al.* (2019)

Expunations

Sun *et al.* (2022)

r/Jokes

Weller and Seppi (2020)





The JokeJudger dataset



- Introduced by Winters *et al.* (2019)
- Contains **521 jokes**, including **100 generated by Goofer**
- A **rating** (ranging **1-5**) is associated to each joke 
- Each joke was rated by a variable number of people, ranging from 4 to 57
- Training a random forest with their metrics, Winters *et al.* (2019) were able to reach an accuracy on rating **prediction of 61.5%**





The JokeJudger dataset



Steps taken:

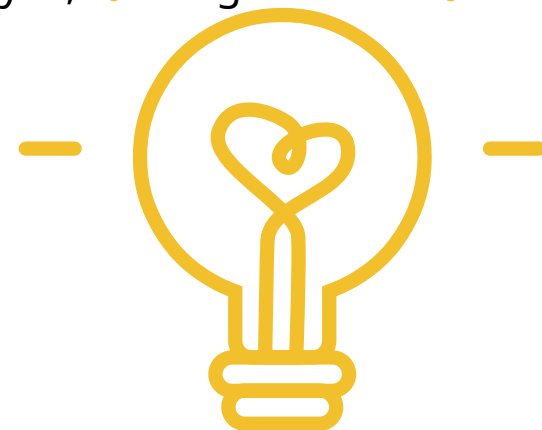
1. The rating was averaged and approximated to a class of 1, 2, 3, 4, or 5 stars.
2. Two sets of embedding vectors, one with MiniLM all-MiniLM-L6-v2 model and one with OpenAI text-embedding-3-small model were calculated for X, Y, Z, XZ, and YZ.
3. Obviousness, compatibility, symmetry and comparison metrics were calculated.
4. With those four metrics and the target rating calculated at step 1, several models were trained: random forest, support vector machine, naive Bayes, and regression.

RESULTS

Best results with random forests:

MiniLM → Accuracy = 57.1%

OpenAI → Accuracy = 54.3%

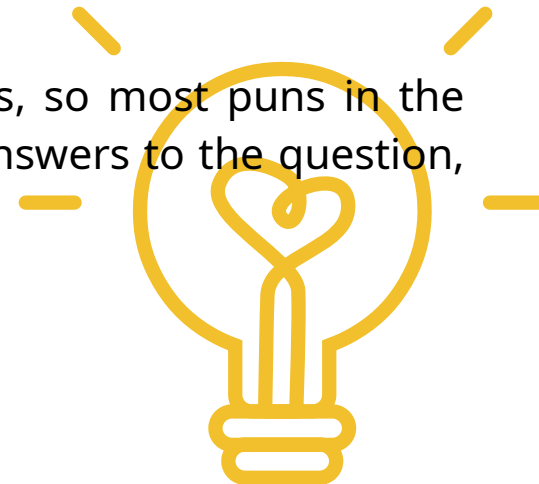




The Expunations dataset



- Introduced by Sun *et al.* (2022) – An annotated selection of puns from the SemEval 2017 Task 7 dataset (Miller *et al.*, 2017)
- Contains **1,999 puns**
- A **rating** (ranging **1-5**) is associated to each joke
- Crowdsourced annotations, including keywords that highlight the most distinctive words contributing to the humor, explanations of why the text is funny, and fine-grained funniness ratings
- Raters were also invited to provide natural language explanations, so most puns in the dataset include at least one explanation. Additionally, it includes answers to the question, “Do you understand the text?”





The Expunations dataset



EXAMPLE



When the glazier was sent to the hospital room to check the cracked window, he told the patient in the body cast, “I’ve come to feel your pane.”

Explanation (included in the dataset): “Pane” is a sheet of glass that a glazier would fit to a window. However, ‘pain’ refers to physical discomfort often felt by a hospital patient.”

Tags: “glazier,” “window,” “feel your pane,” “patient in the body cast,” “hospital,” “cracked window,”





The Expunations dataset



FURTHER EXPANSION OF THE DATASET

Four additional fields were added:

- the expression at the center of the pun in **two versions** (to cover the cases of homophones)
- two sentences that use the expression **in both of its implied meanings**



→ The result was a dataset of 1,895 puns

IN THE PREVIOUS EXAMPLE



- *Expression 1: “pane”*
- *Expression 2: “pain”*
- *Sentence 1: “The glazier installs a pane in the window.”*
- *Sentence 2: “He was in a lot of pain while he was at the hospital.”*





The Expunations dataset



HOW THE DATASET WAS EXPANDED

The content was generated by iteratively querying the OpenAI model (o3-mini-2025-01-31) with the pun, its explanations, keywords, and the following prompt:

Below, you can find a pun, based on the double meaning of an expression, along with it a unique identifier and, if available, some explanations and some keywords.

I want you to:

- select the expression at the core of the joke. If it is a homophone, select the two versions of it.*
- create two serious (without humor) sentences with the same expression: one where it takes the first meaning, and one where it takes the second meaning.*

The result of the computation must be returned in a format like this, to fit a CSV file separated by a semicolon:

ID; expression1; expression2; sentence1; sentence2.

Note: if the expression is a homophone, write the two versions. Otherwise, just write the same expression twice.





The Expunations dataset



THE METRICS REVISITED



Obviousness	Cosine similarity of the embedding vectors of expression 1 (“pane”) and sentence 1 (“ <i>The glazier installs a pane in the window.</i> ”).
Compatibility	Cosine similarity of the embedding vectors of expression 2 (“pain”) and sentence 2 (“ <i>He was in a lot of pain while he was at the hospital.</i> ”). 
Comparison	Difference between two cosine similarities: - on one side the cosine similarity between the text of the pun (“ <i>When the glazier was sent to the hospital room to check the cracked window, he told the patient in the body cast, I’ve come to feel your pane</i> ”) and the sentence 1 (“ <i>The glazier installs a pane in the window.</i> ”); - on the other side, the cosine similarity between the text of the pun and the sentence 2 (“ <i>He was in a lot of pain while he was at the hospital.</i> ”). 
Symmetry	Difference between obviousness and compatibility.





The Expunations dataset



RESULTS

Best results with SVM:

MiniLM → Accuracy = 51.1%

OpenAI → Accuracy = 51.0%

Random forests:

MiniLM → Accuracy = 49.7%

OpenAI → Accuracy = 50.7%

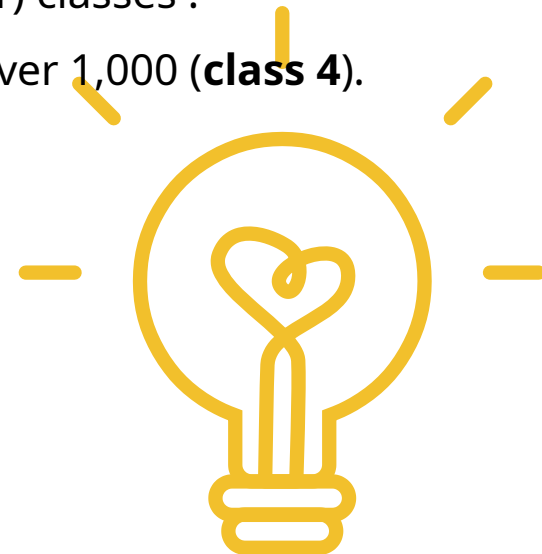




The r/Jokes dataset



- Introduced by Weller and Seppi (2020)
- Contains **550,000 jokes (yes, it's huge)** from the forum Reddit r/Jokes over an 11-year span.
- A **number of upvotes** is associated to each joke 
- The number of upvotes ranges from 1 to several thousand.
- For the purposes of this work, they were mapped into five (non-linear) classes :
0 (**class 0**), 1–10 (**class 1**), 11–100 (**class 2**), 101–1,000 (**class 3**), and over 1,000 (**class 4**). 





The r/Jokes dataset



A selection of jokes was extracted



After a preliminary division of the dataset into several CSV files containing 100 jokes each, the selection was made through an extensive series of requests to the OpenAI model (o3-mini-2025-01-31), using the following prompt:

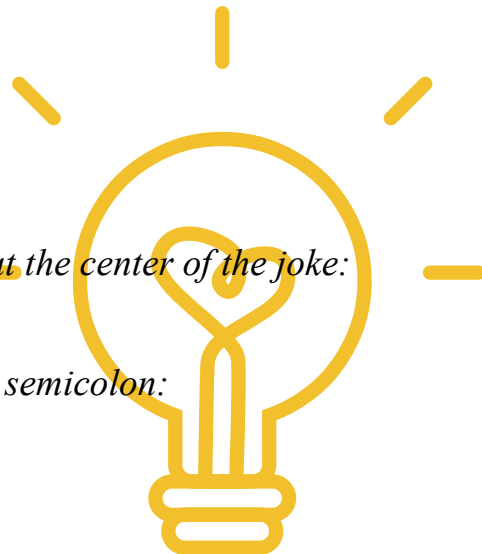
In the csv file, you can find a list of jokes. I want you to select only the ones where the joke is based on a word that has these characteristics:

- In the context of the joke, the word takes two different meanings.*
- It is spelled exactly the same way in both meanings: discard jokes base on homophones.*
- It's a one single word - no composite words.*
- It's a word present in the dictionary.*



For each sentence, then I need you to create two serious (without humor) sentences with the word at the center of the joke: one where the word takes the first meaning, and one where it takes the second meaning.

*The result of the computation must be returned in a format like this, to fit a CSV file separated by a semicolon:
sentence; word; explanation; first new sentence; second new sentence*





The r/Jokes dataset



The jokes (479 were selected) were further filtered based on their structure: only those consisting of two parts, a setup and a punchline, separated by a question mark, were selected and split into two separate fields.



The final dataset included **263 jokes**, formatted as the example that follows.

Setup: *“What did the debater say after getting stabbed during an argument?”*

Punchline: “good point”

Central expression: “point”

Explanation: “A point can refer to a sharp end or tip of an object. It can also mean an argument or assertion in a discussion.”

Sentence 1: “The knife had a sharp point that could easily cut through paper.”

Sentence 2: “During the debate, he made a valid point that swayed the audience.”

Score: 219.0

Score_class: 3





The r/Jokes dataset



THE METRICS REVISITED (AGAIN)



Obviousness	Cosine similarity of the embedding vectors of setup (“What did the debater say after getting stabbed during an argument?”) and sentence 1 (“The knife had a sharp point that could easily cut through paper.”)
Compatibility	Cosine similarity of the embedding vectors of punchline (“good point”) and sentence 2 (“During the debate, he made a valid point that swayed the audience.”)
Comparison	Cosine similarity between the text of the setup and the punchline .
Symmetry	Difference between obviousness and compatibility.





The r/Jokes dataset

RESULTS

Best results with random forests:

SVM → Accuracy = 41.8%

OpenAI → Accuracy = 41.8%



And the winner is

(or the least of the losers)



57.1%

The best performance was achieved on **JokeJudger** dataset, using a **random forest** trained on metrics derived from embedding vectors generated by the **MiniLM** model (*all-MiniLM-L6-v2*).

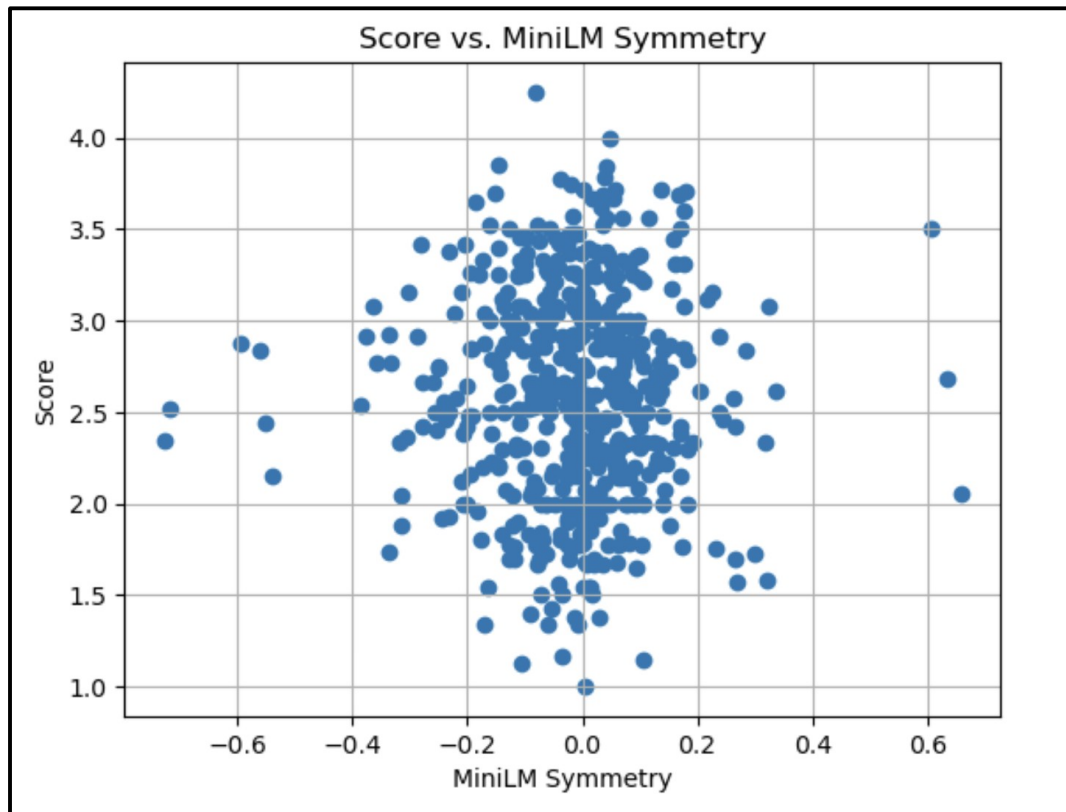
The model reached an **accuracy of 57.1%** on a classification task where scores were discretized into five classes—**falling short of the 61.5% baseline** established by Winters *et al.* (2019).



What happened?



Score vs. Symmetry
(MiniLM model – JokeJugder dataset)



The distribution of the **symmetry** metric shows no clear correlation with humor ratings

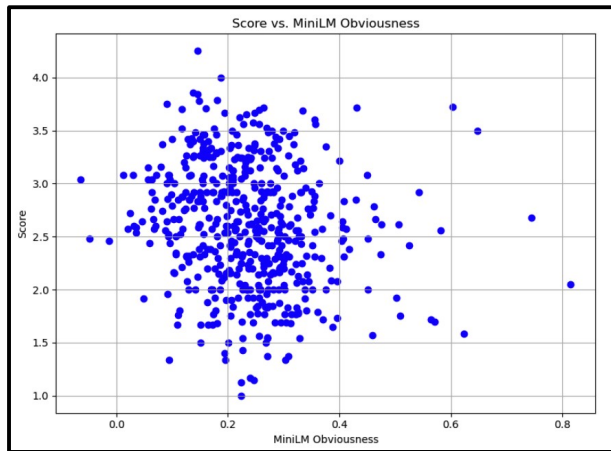




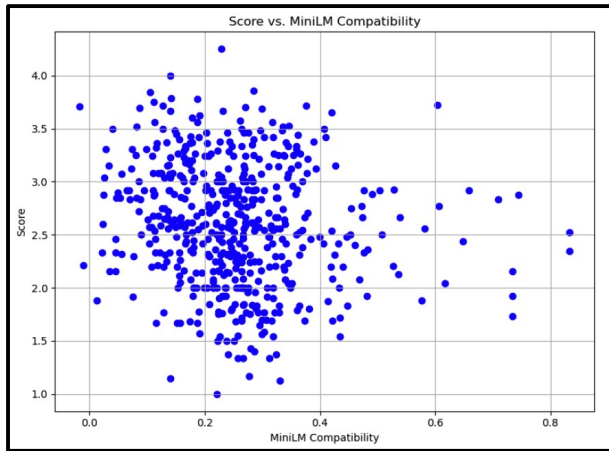
What happened?



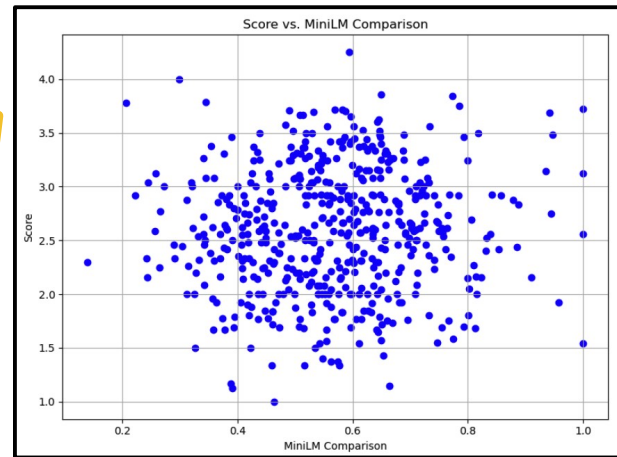
Score vs. Symmetry (MiniLM model – JokeJugder dataset)



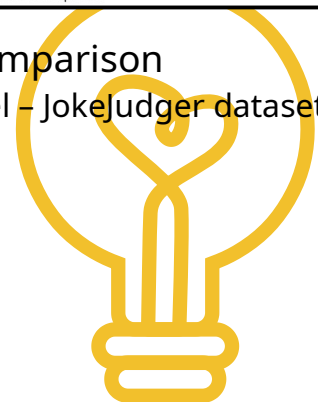
Score vs. Obviousness
(MiniLM model – JokeJugder dataset)



Score vs. Compatibility
(MiniLM model – JokeJugder dataset)



Score vs. Comparison
(MiniLM model – JokeJugder dataset)

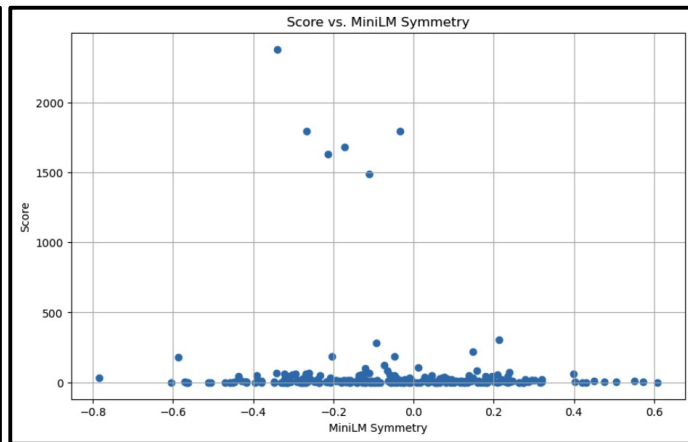
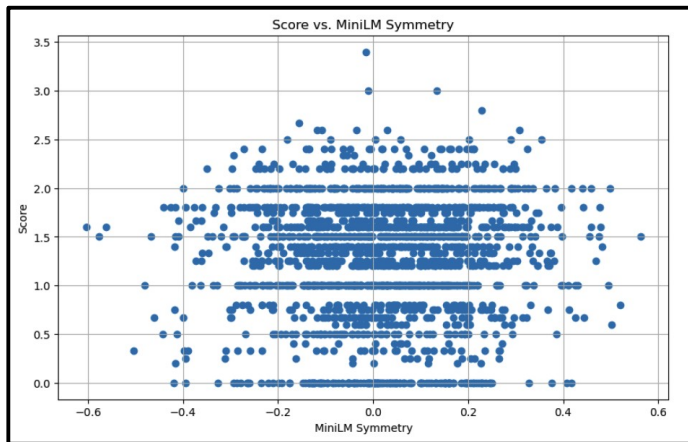
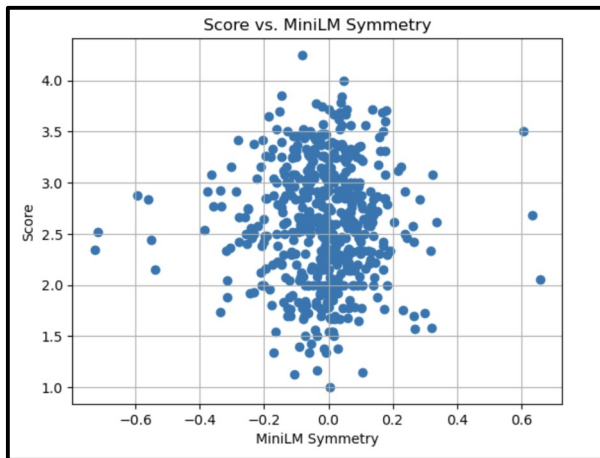




What happened?



Across all three datasets, the highest humor ratings consistently occurred when the symmetry measure was close to zero—indicating that the levels of obviousness and compatibility were nearly equal.



Score vs. Symmetry
(MiniLM model – **JokeJugger** dataset)

Score vs. Symmetry
(MiniLM model – **Expunations** dataset)

Score vs. Symmetry
(MiniLM model – **r/Jokes** dataset)



To the bottom of it



To investigate this further while trying to avoid the issues mentioned above, a subset of puns was manually selected.

Each pun was based on an ambiguous expression that was **both written and pronounced identically**.

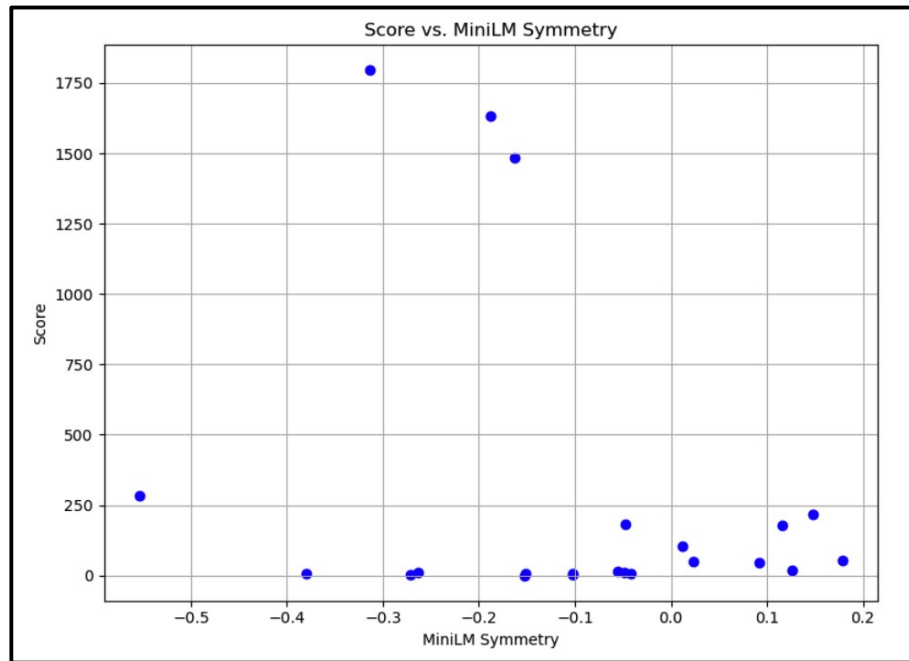
Additionally, **jokes relying on cultural or geographic knowledge were excluded**. The result was a dataset of just 22 jokes. Here are two examples:

“*Why is leather armor best for sneaking? It’s literally made of hide*”

(score: 1486, symmetry: -0.16)

“*What do you call a beaten-up pretzel? A salted pretzel*”

(score: 5, symmetry: -0.04)



Score vs. Symmetry
(MiniLM model – r/Jokes dataset subselection)



Possible explanations

- Most of the jokes in the three datasets are based on **heterographic homophones** expressions that sound the same but differ in both spelling and meaning. For example, “Romance isn’t a science, it’s a heart”
- Some jokes are **not based on wordplay** at all. For instance, in “*How do you stop a lawyer from drowning? Shoot him before he hits the water,*” there is no semantic ambiguity involving a specific word.
- Many jokes draw on **cultural stereotypes** or biases that are unlikely to be fully captured by word embeddings—if they are captured at all.
- Some jokes rely on references to **celebrities** (on top of heterographic homophones), such as: “*What do you call it when Batman skips church? Christian Bale.*” Christian Bale is the actor who portrayed Batman in The Dark Knight trilogy. The punchline plays on the homophonic phrase “Christian bale,” which sounds like “a Christian who bails”—with “bail” being slang for skipping or abandoning something. **The humor in this case depends on both phonetic ambiguity and shared cultural knowledge** about the actor and the character he played.



Possible explanations

- Many jokes also feature **sexual references**. For instance, as Ritchie (1999) points out, the humor in jokes like *"How did the farmer find the sheep in the tall grass? Very satisfying"* is heightened by its inappropriateness. However, as noted earlier, the metric of inappropriateness was not included in the present study.
- **Contextual embeddings** may not be directly comparable using cosine similarity as straightforwardly as this study assumes.
- Using the **OpenAI model** o3-mini-2025-01-31 both **to select jokes** based on certain properties and **to generate sentences** intended to capture their dual meanings may introduce fragility into the system. It is worth noting, however, that this setup—if it has any impact—is more likely to result in overfitting, potentially inflating the performance of the metrics rather than diminishing it.
- The JokeJugder dataset ratings were collected through **an online platform open to anyone interested**, with no random sampling of demographic groups, while **rJokes ratings were extracted as the number of upvoting** a joke collected on the social network Reddit. In both cases, **the average rating may not be representative of the broader population**.



Further work

- Curate a **cleaner dataset** by filtering the Expunations dataset to retain only jokes that rely on wordplay, excluding those based on heterographic homophones, stereotypes, celebrity references, or sexual innuendo. The refined narrowed dataset can then be used to reassess the effectiveness of the four metrics in model training, potentially improving predictive performance.
- Reintroduce a metric for **inappropriateness**, to explore whether the presence of taboo or norm-violating content correlates with perceived humor and contributes to rating predictions.
- Replace machine-generated explanatory sentences with **human-authored sentences**, in order to evaluate whether human interpretation of joke meanings improves metric reliability.
- It is noteworthy –and somewhat surprising– that **earlier studies using simpler techniques**—such as Google Ngrams and WordNet—**have achieved better predictive performance** than those based on embedding methods. A parallel evaluation of the same jokes using both sets of metrics may offer valuable insights into why this was the case.





Why this matters

Large language models such as GPT-4o clearly demonstrate a notable degree of capability in both understanding and generating humor (albeit with limitations that are likely to diminish over time).

Yet, **their most significant limitation lies in their complexity**; we still lack a detailed understanding of how transformers manipulate vector space. Even if they were to exhibit flawless abilities in explaining and generating jokes, the underlying mechanisms—the relative weights and interactions among the elements they process—would remain largely obscure.

Asking ChatGPT to explain a joke is like **asking a cook to prepare a dish**: you may get a result, but the underlying process remains hidden. **Extracting semantic embeddings** and analyzing their relationship to the mechanisms of humor **is more like learning how to cook**.

This is why even if the use of language models is more efficient, the development of simpler techniques is still very relevant to the field.



THANK YOU



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